	Mass of helium gas present $= 0.02407 \times 4.0 = 0.096$ g
18.	Ans: **D** $P = 1/3\ (Nm/V) \langle c^2 \rangle = 1/3\ \rho \langle c^2 \rangle$ $\langle c^2 \rangle = (1.00 \times 10^5)(3) / 1.60 \Rightarrow$ root mean square speed $= \sqrt{\langle c^2 \rangle} = 433\ ms^{-1}$